

Physics Publications with Major Personal Contributions

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October 02, 2026

1. Search for dark matter produced in association with a low-multiplicity jet in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2610.00995](#) | 2026-10-01

2. Measurement of the triple-differential cross section of Z+jet production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2609.39653](#) | 2026-09-30

3. Search for resonant and nonresonant production of pairs of dijet resonances with b jets in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2609.21850](#) | 2026-09-18

4. Calibration of electromagnetic shower features in the CMS calorimeter with machine-learning techniques

Gevorgyan, Arzunik et al.

[arXiv:2609.15377](#) | 2026-09-14

5. Search for a light pseudoscalar Higgs boson in final states with boosted muon pairs and tau lepton pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2609.12683](#) | 2026-09-11

6. Search for the rare Higgs boson decay $H \rightarrow Z\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ and 13.6 TeV

Gevorgyan, Arzunik et al.

[arXiv:2609.04402](#) | 2026-09-03

7. Spin-dependent azimuthal asymmetry of coherent J/ψ photoproduction in hadronic PbPb collisions at $\sqrt{s_{NN}} = 5.36$ TeV

Belyaev, Andrey et al.

[arXiv:2609.04376](#) | 2026-09-03

8. Constraint on the Higgs boson width in the diphoton final state using signal-background interference in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2609.01385](#) | 2026-09-01

9. Evidence for the dead-cone effect in bottom quark initiated jets produced in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.29412](#) | 2026-08-29

10. Search for scalar leptoquarks produced via muon-quark scattering in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.28377](#) | 2026-08-28

11. Search for top quark flavor-changing neutral currents in multilepton final states using an effective field theory approach in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.19547](#) | 2026-08-20

12. Observation of an excess at the top quark pair production threshold in the single-lepton channel

Gevorgyan, Arzunik et al.

[arXiv:2608.19546](#) | 2026-08-20

13. Performance of heavy-flavour jet identification in the CMS high-level trigger in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.18331](#) | 2026-08-18

14. Search for anomalous couplings in WW and WZ production with single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.18245](#) | 2026-08-18

15. Search for a resonance in events with four top quarks decaying into two leptons and jets in proton-proton collisions

Gevorgyan, Arzunik et al.

[arXiv:2608.10148](#) | 2026-08-10

16. Search for resonant production of lepton-enriched semivisible jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.05323](#) | 2026-08-05

17. Combination of vector boson scattering measurements in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2608.01289](#) | 2026-08-02

18. Search for a top-philic Z' boson decaying into a $t\bar{t}$ pair in a final state with jets and an electron or muon in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2608.01251](#) | 2026-08-02

19. A measurement of the Higgs boson mass in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.28396](#) | 2026-07-30

20. Measurement of the fragmentation properties of jets containing $Y(nS)$ mesons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.25653](#) | 2026-07-28

21. Off-shell Higgs boson measurements: Yukawa couplings, self-coupling, compositeness, and width

Gevorgyan, Arzunik et al.

[arXiv:2607.23352](#) | 2026-07-25

22. Measurement of the $\gamma\gamma \rightarrow \tau\tau$ cross section and constraints on the anomalous magnetic moment of the τ lepton in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2607.20148](#) | 2026-07-22

23. Study of χ_c production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Belyaev, Andrey et al.

[arXiv:2607.17847](#) | 2026-07-20

24. Evidence for sequential $\Upsilon(nS)$ suppression in light ion collisions

Belyaev, Andrey et al.

[arXiv:2607.13758](#) | 2026-07-15

25. Exploring the origin of D^0 meson elliptic flow in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using event shape engineering

Gevorgyan, Arzunik et al.

[arXiv:2607.11017](#) | 2026-07-13

26. Search for dark matter in a signature with a four-prong large-radius jet in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.11016](#) | 2026-07-13

27. Search for soft unclustered energy patterns in proton-proton collisions at $\sqrt{s} = 13$ TeV using data scouting

Gevorgyan, Arzunik et al.

[arXiv:2607.09621](#) | 2026-07-10

28. Search for nonresonant triple Higgs boson production in the final state with six bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.05145](#) | 2026-07-06

29. Study of ZZ and ZH production in the $b\bar{b}\tau\tau$ final state and search for high-mass spin-0 and spin-1 resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2607.01589](#) | 2026-07-02

30. Search for physics beyond the standard model in four and three top quark production events using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2606.30862](#) | 2026-06-29

31. Precision luminosity measurement in proton-proton collisions at a center-of-mass energy of 13 TeV with the CMS detector at the Large Hadron Collider

Gevorgyan, Arzunik et al.

[arXiv:2606.26832](#) | 2026-06-25

32. Search for long-lived particles decaying into muons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV using data scouting

Gevorgyan, Arzunik et al.

[arXiv:2606.25840](#) | 2026-06-24

33. Observation of electroweak production of pairs of Z bosons in proton-proton collisions at 13 TeV

Gevorgyan, Arzunik et al.

[arXiv:2606.20558](#) | 2026-06-18

34. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and tau leptons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2606.03510](#) | 2026-06-02

35. Search for single production of a vector-like B' quark decaying to a top quark and a W boson in the single-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tauqeer, Komal et al.

[arXiv:2606.01423](#) | 2026-05-31

36. Dependence of two-particle azimuthal correlations on the forward rapidity gap width in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Hayrapetyan, Aram et al.

[arXiv:2606.00171](#) | 2026-05-29

37. Search for light scalar particles produced in Higgs boson decays in exclusive final states with two muons and two hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.27875](#) | 2026-05-27

38. Search for a new heavy scalar resonance decaying to a pair of Z bosons in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.26462](#) | 2026-05-26

39. Search for low-mass resonances decaying to $\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.25103](#) | 2026-05-24

40. Studies of $Z \rightarrow 4\ell$ decays in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Tumasyan, Armen et al.

[arXiv:2605.25158](#) | 2026-05-24

41. Search for Higgs boson decays into two neutral scalars with unequal masses in final states with b quarks and tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.19144](#) | 2026-05-18

42. Search for the Nonresonant Production of a Pair of Additional Higgs Bosons in the Type-X Two-Higgs-Doublet Model in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 137 \(2026\) 131801](#) | [arXiv:2605.16655](#) | [DOI:10.1103/qydl-rlp7](#) | 2026-05-15

43. Search for new physics in triple boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV using the effective field theory approach

Hayrapetyan, Aram et al.

[arXiv:2605.15023](#) | 2026-05-14

44. Constraints on anomalous Higgs boson couplings to vector bosons and fermions using the $\gamma\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.14245](#) | 2026-05-14

45. First measurements of vector boson scattering in $W^{\pm}W^{\pm}$ and WZ production in all-leptonic final states at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.15396](#) | 2026-05-14

46. Search for pair production of additional neutral scalars within the Inert Doublet Model in a final state with two electrons or two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.13614](#) | 2026-05-13

47. Search for single vector-like quark production in opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

JHEP 09 (2026) 230 | [arXiv:2605.13649](#) | [DOI:10.1007/JHEP09\(2026\)230](#) | 2026-05-13

48. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

49. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

50. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

JHEP 09 (2026) 182 | [arXiv:2604.27795](#) | [DOI:10.1007/JHEP09\(2026\)182](#) | 2026-05-01

51. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

52. Measurement of the top quark pair production cross section in PbPb collisions at $\sqrt{s_{NN}}=5.36$ TeV

Belyaev, Andrey et al.

JHEP 08 (2026) 054 | [arXiv:2604.27091](#) | [DOI:10.1007/JHEP08\(2026\)054](#) | 2026-04-29

53. Search for Electroweakinos in Compressed-Spectrum Scenarios with Low-Momentum Isolated Tracks in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.Lett. 137 (2026) 131802 | [arXiv:2604.25604](#) | [DOI:10.1103/g4ym-rzkt](#) | 2026-04-28

54. Simultaneous measurements of N -subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.

JHEP 09 (2026) 258 | [arXiv:2604.25538](#) | [DOI:10.1007/JHEP09\(2026\)258](#) | 2026-04-28

55. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

56. Search for Associated Production of a Higgs Boson and Two Vector Bosons via Vector-Boson Scattering at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.Lett. 137 (2026) 101806 | [arXiv:2604.24531](#) | [DOI:10.1103/gwj9-5bxm](#) | 2026-04-27

57. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

58. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

59. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, A. et al.

Phys.Rev.D 114 (2026) 052004 | [arXiv:2604.13320](#) | [DOI:10.1103/hb72-vjj6](#) | 2026-04-14

60. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

61. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

JHEP 08 (2026) 098 | [arXiv:2604.10732](#) | [DOI:10.1007/JHEP08\(2026\)098](#) | 2026-04-12

62. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Lett.B 880 (2026) 140812 | [arXiv:2604.10729](#) | [DOI:10.1016/j.physletb.2026.140812](#) | 2026-04-12

63. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

64. Observation of Nuclear Suppression in Coherent $\pi(1S)$ Photoproduction off Heavy Nuclei at the LHC

Belyaev, Andrey et al.

Phys.Rev.Lett. 137 (2026) 142301 | [arXiv:2604.05814](#) | [DOI:10.1103/l13f-l5xv](#) | 2026-04-07

65. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Lett.B 879 (2026) 140671 | [arXiv:2604.05996](#) | [DOI:10.1016/j.physletb.2026.140671](#) | 2026-04-07

66. Evidence of $ZZ\gamma$ Production and Observation of $4\ell\gamma$ in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.Lett. 137 (2026) 091805 | [arXiv:2604.02594](#) | [DOI:10.1103/gdbv-wnpp](#) | 2026-04-03

67. Search for Higgs boson pair production in the $bbWW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s}=13.6$ TeV

Gevorgyan, Arzunik et al.

JHEP 08 (2026) 019 | [arXiv:2604.02127](#) | [DOI:10.1007/JHEP08\(2026\)019](#) | 2026-04-02

68. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

69. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2603.23454](#) | 2026-03-24

70. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.21965](#) | 2026-03-23

71. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 07 \(2026\) 248](#) | [arXiv:2603.22233](#) | [DOI:10.1007/JHEP07\(2026\)248](#) | 2026-03-23

72. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.
[JHEP 07 \(2026\) 054](#) | [arXiv:2603.19963](#) | [DOI:10.1007/JHEP07\(2026\)054](#) | 2026-03-20

73. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.
[arXiv:2603.14385](#) | 2026-03-15

74. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s}=5.02$ TeV

Hayrapetyan, Aram et al.
[JHEP 08 \(2026\) 223](#) | [arXiv:2603.13592](#) | [DOI:10.1007/JHEP08\(2026\)223](#) | 2026-03-13

75. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.11035](#) | 2026-03-11

76. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.03689](#) | 2026-03-04

77. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[Phys.Rev.D 114 \(2026\) 032005](#) | [arXiv:2603.03250](#) | [DOI:10.1103/q2mh-7mhj](#) | 2026-03-03

78. Combination of ATLAS and CMS Searches for Higgs Boson Pair Production at $\sqrt{s}=13$ TeV

Aad, Georges et al.
[Phys.Rev.Lett. 137 \(2026\) 131803](#) | [arXiv:2602.23991](#) | [DOI:10.1103/27s7-2ftr](#) | 2026-02-27

79. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.
[Phys.Lett.B 880 \(2026\) 140679](#) | [arXiv:2602.21325](#) | [DOI:10.1016/j.physletb.2026.140679](#) | 2026-02-24

80. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[Phys.Rev.D 113 \(2026\) 112005](#) | [arXiv:2602.20477](#) | [DOI:10.1103/bwqs-vgqn](#) | 2026-02-24

81. Observation of the Jet Diffusion Wake Using Dijets in Heavy-Ion Collisions

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 137 \(2026\) 071902](#) | [arXiv:2602.19431](#) | [DOI:10.1103/g49y-8cjl](#) | 2026-02-23

82. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

83. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

84. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 89 \(2026\) 087801](#) | [arXiv:2602.18611](#) | [DOI:10.1088/1361-6633/ae72c4](#) | 2026-02-20

85. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

86. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

87. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[JINST 21 \(2026\) P04002](#) | [arXiv:2602.11359](#) | [DOI:10.1088/1748-0221/21/04/P04002](#) | 2026-02-11

88. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[Phys.Rev.D 114 \(2026\) 012005](#) | [arXiv:2602.10270](#) | [DOI:10.1103/36cv-vst3](#) | 2026-02-10

89. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

90. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

91. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

92. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $bb4q$ final state at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2026\) 050](#) | [arXiv:2602.00273](#) | [DOI:10.1007/JHEP08\(2026\)050](#) | 2026-01-30

93. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2026\) 187](#) | [arXiv:2601.21574](#) | [DOI:10.1007/JHEP06\(2026\)187](#) | 2026-01-29

94. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s}=13.6$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 106](#) | [arXiv:2601.20023](#) | [DOI:10.1007/JHEP07\(2026\)106](#) | 2026-01-27

95. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 86 \(2026\) 901](#) | [arXiv:2601.20011](#) | [DOI:10.1140/epjc/s10052-026-15699-9](#) | 2026-01-27

96. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 878 \(2026\) 140498](#) | [arXiv:2601.20063](#) | [DOI:10.1016/j.physletb.2026.140498](#) | 2026-01-27

97. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s}=13.6$ TeV

Hayrapetyan, Aram et al.

[Phys.Rept. 1205 \(2027\) 2456](#) | [arXiv:2601.17544](#) | [DOI:10.1016/j.physrep.2026.07.007](#) | 2026-01-24

98. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 102](#) | [arXiv:2601.17576](#) | [DOI:10.1007/JHEP07\(2026\)102](#) | 2026-01-24

99. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 86 \(2026\) 1025](#) | [arXiv:2601.17554](#) | [DOI:10.1140/epjc/s10052-026-15754-5](#) | 2026-01-24

100. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2026\) 049](#) | [arXiv:2601.14102](#) | [DOI:10.1007/JHEP08\(2026\)049](#) | 2026-01-20

101. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2026\) 005](#) | [arXiv:2601.12583](#) | [DOI:10.1007/JHEP07\(2026\)005](#) | 2026-01-18

102. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092016](#) | [arXiv:2601.11330](#) | [DOI:10.1103/2m87-3dlr](#) | 2026-01-16

103. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

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Chekhovsky, Vladimir et al.

[JHEP 05 \(2025\) 079](#) | [arXiv:2501.14849](#) | [DOI:10.1007/JHEP05\(2025\)079](#) | 2025-01-24

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Hayrapetyan, Aram et al.

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 061803](#) | [arXiv:2412.19261](#) | [DOI:10.1103/srvm-flh3](#) | 2024-12-26

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Chekhovsky, Vladimir et al.

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Hayrapetyan, Aram et al.

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Hayrapetyan, Aram et al.

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Chekhovsky, Vladimir et al.

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[Phys.Rev.D 111 \(2025\) 112004](#) | [arXiv:2412.04357](#) | [DOI:10.1103/PhysRevD.111.112004](#) | 2024-12-05

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Hayrapetyan, Aram et al.

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[Phys.Lett.B 866 \(2025\) 139568](#) | [arXiv:2412.00570](#) | [DOI:10.1016/j.physletb.2025.139568](#) | 2024-11-30

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205. Search for rare decays of the Z and Higgs bosons to a J/ψ or $\psi(2S)$ meson and a photon in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

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[JHEP 06 \(2025\) 006](#) | [arXiv:2410.21137](#) | [DOI:10.1007/JHEP06\(2025\)006](#) | 2024-10-28

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[JINST 19 \(2024\) P11021](#) | [arXiv:2410.17038](#) | [DOI:10.1088/1748-0221/19/11/P11021](#) | 2024-10-22

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[JHEP 04 \(2025\) 109](#) | [arXiv:2410.09164](#) | [DOI:10.1007/JHEP04\(2025\)109](#) | 2024-10-11

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[Phys.Rev.D 110 \(2024\) 112016](#) | [arXiv:2409.11067](#) | [DOI:10.1103/PhysRevD.110.112016](#) | 2024-09-17

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[Phys.Lett.B 866 \(2025\) 139526](#) | [arXiv:2408.07622](#) | [DOI:10.1016/j.physletb.2025.139526](#) | 2024-08-14

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[Phys.Rev.D 111 \(2025\) L051102](#) | [arXiv:2408.00678](#) | [DOI:10.1103/PhysRevD.111.L051102](#) | 2024-08-01

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[Phys.Lett.B 861 \(2025\) 139210](#) | [arXiv:2407.13554](#) | [DOI:10.1016/j.physletb.2024.139210](#) | 2024-07-18

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[JHEP 02 \(2025\) 036](#) | [arXiv:2407.10717](#) | [DOI:10.1007/JHEP02\(2025\)036](#) | 2024-07-15

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[Phys.Rev.D 110 \(2024\) 092002](#) | [arXiv:2407.03223](#) | [DOI:10.1103/PhysRevD.110.092002](#) | 2024-07-03

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[Phys.Rept. 1115 \(2025\) 3](#) | [arXiv:2405.18661](#) | [DOI:10.1016/j.physrep.2024.11.005](#) | 2024-05-28

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250. Study of WH production through vector boson scattering and extraction of the relative sign of the W and Z couplings to the Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 857 \(2024\) 138979](#) | [arXiv:2405.14757](#) | [DOI:10.1016/j.physletb.2024.138979](#) | 2024-05-23

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254. Overview of high-density QCD studies with the CMS experiment at the LHC

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[Phys.Rept. 1115 \(2025\) 2367](#) | [arXiv:2405.10785](#) | [DOI:10.1016/j.physrep.2024.11.007](#) | 2024-05-17

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259. Search for the Z Boson Decay to $\tau\tau\mu\mu$ in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

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[JINST 19 \(2024\) P09012](#) | [arXiv:2404.17377](#) | [DOI:10.1088/1748-0221/19/09/P09012](#) | 2024-04-26

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[Eur.Phys.J.C 84 \(2024\) 842](#) | [arXiv:2404.16082](#) | [DOI:10.1140/epjc/s10052-024-13116-7](#) | 2024-04-24

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[JHEP 10 \(2024\) 209](#) | [arXiv:2404.02711](#) | [DOI:10.1007/JHEP10\(2024\)209](#) | 2024-04-03

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[Phys.Rev.Lett. 133 \(2024\) 201803](#) | [arXiv:2404.02992](#) | [DOI:10.1103/PhysRevLett.133.201803](#) | 2024-04-03

266. Measurement of the production cross section of a Higgs boson with large transverse momentum in its decays to a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Lett.B 857 \(2024\) 138964](#) | [arXiv:2403.20201](#) | [DOI:10.1016/j.physletb.2024.138964](#) | 2024-03-29

267. Search for ZZ and ZH production in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Eur.Phys.J.C 84 \(2024\) 712](#) | [arXiv:2403.20241](#) | [DOI:10.1140/epjc/s10052-024-13021-z](#) | 2024-03-29

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Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2342](#) | [arXiv:2403.16926](#) | [DOI:10.1016/j.physrep.2024.09.004](#) | 2024-03-25

269. Enriching the physics program of the CMS experiment via data scouting and data parking

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2344](#) | [arXiv:2403.16134](#) | [DOI:10.1016/j.physrep.2024.09.006](#) | 2024-03-24

270. Performance of the CMS electromagnetic calorimeter in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P09004](#) | [arXiv:2403.15518](#) | [DOI:10.1088/1748-0221/19/09/P09004](#) | 2024-03-22

271. Observation of the $J/\psi \rightarrow \mu^+ \mu^- \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) L111101](#) | [arXiv:2403.11352](#) | [DOI:10.1103/PhysRevD.109.L111101](#) | 2024-03-17

272. Search for the decay of the Higgs boson to a pair of light pseudoscalar bosons in the final state with four bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 097](#) | [arXiv:2403.10341](#) | [DOI:10.1007/JHEP06\(2024\)097](#) | 2024-03-15

273. Search for Higgs boson pair production in the bbW^+W^- decay mode in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2024\) 293](#) | [arXiv:2403.09430](#) | [DOI:10.1007/JHEP07\(2024\)293](#) | 2024-03-14

274. Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 191902](#) | [arXiv:2403.05311](#) | [DOI:10.1103/PhysRevLett.133.191902](#) | 2024-03-08

275. Search for long-lived heavy neutrinos in the decays of B mesons produced in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 183](#) | [arXiv:2403.04584](#) | [DOI:10.1007/JHEP06\(2024\)183](#) | 2024-03-07

276. Search for dark QCD with emerging jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2024\) 142](#) | [arXiv:2403.01556](#) | [DOI:10.1007/JHEP07\(2024\)142](#) | 2024-03-03

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Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2369](#) | [arXiv:2403.01313](#) | [DOI:10.1016/j.physrep.2024.12.002](#) | 2024-03-02

278. Constraints on anomalous Higgs boson couplings from its production and decay using the WW channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 779](#) | [arXiv:2403.00657](#) | [DOI:10.1140/epjc/s10052-024-12925-0](#) | 2024-03-01

279. Search for heavy neutral leptons in final states with electrons, muons, and hadronically decaying tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

280. Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

281. Search for baryon number violation in top quark production and decay using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

282. Observation of the $\Xi_b^- \rightarrow \psi(2S) \Xi^-$ decay and studies of the $\Xi_b(5945)^0$ baryon in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

283. Search for long-lived particles using displaced vertices and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

284. Portable Acceleration of CMS Computing Workflows with Coprocessors as a Service

Hayrapetyan, Aram et al.

[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

285. Search for long-lived particles decaying to final states with a pair of muons in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

286. Measurement of Energy Correlators inside Jets and Determination of the Strong Coupling $\alpha_S(m_Z)$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 071903](#) | [arXiv:2402.13864](#) | [DOI:10.1103/PhysRevLett.133.071903](#) | 2024-02-21

287. Search for bottom-type vectorlike quark pair production in dileptonic and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

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Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

289. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

290. Search for Fractionally Charged Particles in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

292. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

293. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

294. Search for long-lived particles decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

295. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

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Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 1062](#) | [arXiv:2401.16303](#) | [DOI:10.1140/epjc/s10052-024-13114-9](#) | 2024-01-29

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Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

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Hayrapetyan, Aram et al.

[JHEP 01 \(2025\) 011](#) | [arXiv:2401.11355](#) | [DOI:10.1007/JHEP01\(2025\)011](#) | 2024-01-20

299. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$ and $B^{\pm} \rightarrow K^{\pm} e^+ e^-$ decays in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

300. Extracting the speed of sound in quark-gluon plasma with ultrarelativistic lead-lead collisions at the LHC

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[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 142301](#) | [arXiv:2312.17103](#) | [DOI:10.1103/PhysRevLett.133.142301](#) | 2023-12-28

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[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

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Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

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[JHEP 05 \(2024\) 116](#) | [arXiv:2312.16343](#) | [DOI:10.1007/JHEP05\(2024\)116](#) | 2023-12-26

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Hayrapetyan, Aram et al.

[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

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Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

307. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

308. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H\rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

309. Search for charged-lepton flavor violation in the production and decay of top quarks using trilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 012009](#) | [arXiv:2312.03199](#) | [DOI:10.1103/PhysRevD.111.012009](#) | 2023-12-05

310. Search for the lepton flavor violating $\tau \rightarrow 3\mu$ decay in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

311. Higher-order moments of the elliptic flow distribution in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

312. Search for new Higgs bosons via same-sign top quark pair production in association with a jet in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

313. Search for high-mass exclusive diphoton production with tagged protons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

314. Search for an exotic decay of the Higgs boson into a Z boson and a pseudoscalar particle in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

315. Search for W' bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

317. Search for dark matter particles in W^+W^- events with transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-10-18

318. Search for central exclusive production of top quark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV with tagged protons

Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

319. Observation of $WW\gamma$ production and search for $H\gamma$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

320. Muon identification using multivariate techniques in the CMS experiment in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

321. Study of azimuthal anisotropy of $\phi(1S)$ mesons in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

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[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

322. Search for stealth supersymmetry in final states with two photons, jets, and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

323. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

324. Search for supersymmetry in final states with disappearing tracks in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

325. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

326. Inclusive and differential cross section measurements of $t\bar{t}b\bar{b}$ production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

327. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

328. Development of the CMS detector for the CERN LHC Run 3

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

329. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC

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[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

330. Luminosity determination using Z boson production at the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

331. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

332. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

333. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

334. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

335. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

336. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

337. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

338. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

339. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

340. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

341. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

342. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

Tumasyan, Armen et al.

[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

343. New Structures in the $J/\psi/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12

344. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29

345. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26

346. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22

347. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19

348. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13

349. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12

350. Observation of the rare decay of the η meson to four muons

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08

351. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14

352. Probing Small Bjorken-x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29

353. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29

354. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

355. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08

356. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07

357. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02

358. Search for a vector-like quark $T' \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24

359. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03

360. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19

361. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18

362. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12

363. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, A. et al.

[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23

364. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23

365. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

366. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

367. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

368. Measurements of azimuthal anisotropy of nonprompt D0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

369. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

370. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

371. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

372. Measurement of the differential $t\bar{t}$ production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

373. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

374. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

375. Application of Portable Parallelization Strategies for GPUs on track reconstruction kernels

Kwok, Martin et al.

376. Search for medium effects using jets from bottom quarks in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

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JHEP 07 (2023) 139 | arXiv:2210.08325 | DOI:10.1007/JHEP07(2023)139 | 2022-10-15

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379. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

Phys.Lett.B 843 (2023) 137803 | arXiv:2210.03082 | DOI:10.1016/j.physletb.2023.137803 | 2022-10-06

380. Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

Phys.Lett.B 844 (2023) 137813 | arXiv:2210.00043 | DOI:10.1016/j.physletb.2023.137813 | 2022-09-30

381. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

JHEP 07 (2023) 020 | arXiv:2209.07327 | DOI:10.1007/JHEP07(2023)020 | 2022-09-15

382. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

Phys.Rev.Lett. 131 (2023) 101801 | arXiv:2209.06197 | DOI:10.1103/PhysRevLett.131.101801 | 2022-09-13

383. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

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384. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

JHEP 07 (2023) 091 | arXiv:2208.12279 | DOI:10.1007/JHEP07(2023)091 | 2022-08-25

385. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

Phys.Lett.B 846 (2023) 137713 | arXiv:2208.09700 | DOI:10.1016/j.physletb.2023.137713 | 2022-08-20

386. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

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387. Search for CP violation in ttH and tH production in multilepton channels in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

388. Measurement of the $t\bar{t}$ charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

389. Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04

390. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

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Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

392. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

393. Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 067](#) | [arXiv:2202.06075](#) | [DOI:10.1007/JHEP07\(2022\)067](#) | 2022-07-11

394. Measurement of the top quark pole mass using $t\bar{t}$ +jet events in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

395. Search for direct pair production of supersymmetric partners of τ leptons in the final state with two hadronically decaying τ leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

396. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 032](#) | [arXiv:2207.01046](#) | [DOI:10.1007/JHEP09\(2023\)032](#) | 2022-07-03

397. Search for Higgs boson pairs decaying to WW^*WW^* , $WW^*\tau\tau$, and $\tau\tau\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

398. Search for nonresonant Higgs boson pair production in the four leptons plus twob jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

399. Search for resonant and nonresonant production of pairs of dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 161](#) | [arXiv:2206.09997](#) | [DOI:10.1007/JHEP07\(2023\)161](#) | 2022-06-20

400. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

401. Measurements of the Higgs boson production cross section and couplings in the W boson pair decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19

402. Probing Heavy Majorana Neutrinos and the Weinberg Operator through Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

403. Precision measurement of the Z boson invisible width in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14

404. Observation of τ lepton pair production in ultraperipheral lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10

405. Search for Higgs boson decays into Z and J/ψ and for Higgs and Z boson decays into J/ψ or Y pairs in pp collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07

406. Observation of same-sign WW production from double parton scattering in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06

407. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s} = 7$ and 8 TeV

Aad, G. et al.

[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27

408. Search for electroweak production of charginos and neutralinos at $\sqrt{s}=13$ TeV in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19

409. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17

410. Search for CP violating top quark couplings in pp collisions at $\sqrt{s} = 13$ TeV

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[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15

411. Search for heavy resonances and quantum black holes in $e\mu$, $e\tau$, and $\mu\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 227](#) | [arXiv:2205.06709](#) | [DOI:10.1007/JHEP05\(2023\)227](#) | 2022-05-13

412. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13

413. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 061801](#) | [arXiv:2205.05550](#) | [DOI:10.1103/PhysRevLett.131.061801](#) | 2022-05-11

414. Observation of electroweak $W+W^-$ pair production in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

415. Measurement of the mass dependence of the transverse momentum of lepton pairs in Drell-Yan production in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 83 \(2023\) 628](#) | [arXiv:2205.04897](#) | [DOI:10.1140/epjc/s10052-023-11631-7](#) | 2022-05-10

416. Constraints on anomalous Higgs boson couplings to vector bosons and fermions from the production of Higgs bosons using the $\tau\tau$ final state

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032013](#) | [arXiv:2205.05120](#) | [DOI:10.1103/PhysRevD.108.032013](#) | 2022-05-10

417. CMS pythia 8 colour reconnection tunes based on underlying-event data

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

418. Measurement of differential cross sections for the production of a Z boson in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

419. Search for CP violation using $t\bar{t}$ events in the lepton+jets channel in pp collisions at $\sqrt{s} = 13$ TeV

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[JHEP 06 \(2023\) 081](#) | [arXiv:2205.02314](#) | [DOI:10.1007/JHEP06\(2023\)081](#) | 2022-05-04

420. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012009](#) | [arXiv:2205.01835](#) | [DOI:10.1103/PhysRevD.108.012009](#) | 2022-05-03

421. Strange hadron collectivity in pPb and PbPb collisions

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[JHEP 05 \(2023\) 007](#) | [arXiv:2205.00080](#) | [DOI:10.1007/JHEP05\(2023\)007](#) | 2022-04-29

422. Azimuthal Correlations within Exclusive Dijets with Large Momentum Transfer in Photon-Lead Collisions

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

423. Two-particle azimuthal correlations in $\gamma\gamma$ interactions using pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV
Tumasyan, Armen et al.
[Phys.Lett.B 844 \(2023\) 137905](#) | [arXiv:2204.13486](#) | [DOI:10.1016/j.physletb.2023.137905](#) | 2022-04-28
424. Search for light Higgs bosons from supersymmetric cascade decays in pp collisions at $\sqrt{s}=13$ TeV
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28
425. Search for Higgs boson decays to a Z boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[JHEP 05 \(2023\) 233](#) | [arXiv:2204.12945](#) | [DOI:10.1007/JHEP05\(2023\)233](#) | 2022-04-27
426. Measurements of Higgs boson production in the decay channel with a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 562](#) | [arXiv:2204.12957](#) | [DOI:10.1140/epjc/s10052-023-11452-8](#) | 2022-04-27
427. Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the four b quarks final state with boosted topology
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438. Search for resonances decaying to three W bosons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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439. Identification of hadronic tau lepton decays using a deep neural network

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[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

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[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

441. Search for charged-lepton flavor violation in top quark production and decay in pp collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

442. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the dilepton channel and effective field theory interpretation in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

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444. Search for higgsinos decaying to two Higgs bosons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

445. Observation of the B_c^+ Meson in Pb-Pb and pp Collisions at $\sqrt{s_{NN}}=5.02 \text{ TeV}$ and Measurement of its Nuclear Modification Factor

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[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

446. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

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[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

447. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

448. Search for long-lived particles decaying into muon pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV collected with a dedicated high-rate data stream

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[JHEP 04 \(2022\) 062](#) | [arXiv:2112.13769](#) | [DOI:10.1007/JHEP04\(2022\)062](#) | 2021-12-27

449. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

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[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

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Tumasyan, Armen et al.

[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

452. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

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[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

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[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

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[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

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[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

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[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

458. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s} = 8$ TeV

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[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

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Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

460. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

461. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

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Tumasyan, Armen et al.

[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

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[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

464. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

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[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

466. Study of dijet events with large rapidity separation in proton-proton collisions at $\sqrt{s} = 2.76$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

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Tumasyan, Armen et al.

[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

468. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

469. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, A. et al.

[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

470. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

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Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

472. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

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[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

474. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

475. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

476. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

477. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

478. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

479. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

480. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

481. Observation of tW production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

482. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

483. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

484. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

485. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

486. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

487. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

488. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

489. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

490. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

491. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

492. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

493. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

494. Fragmentation of jets containing a prompt J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

495. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

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Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

497. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

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[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

498. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

499. Measurements of the $pp \rightarrow W^{\pm}\gamma\gamma$ and $pp \rightarrow Z\gamma\gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

500. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

501. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

502. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

503. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

504. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

505. Search for W' bosons decaying to a top and a bottom quark at $\sqrt{s}=13\text{TeV}$ in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

506. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

507. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$ in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

508. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{\text{NN}}}=5.02 \text{ TeV}$

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[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

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[JINST 14 \(2019\) P07004](#) | [arXiv:1903.06078](#) | [DOI:10.1088/1748-0221/14/07/P07004](#) | 2019-03-14

675. Search for charged Higgs bosons in the $H^\pm \rightarrow \tau^\pm \nu_\tau$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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676. Measurement of electroweak production of a W boson in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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Sirunyan, Albert M et al.

[JINST 14 \(2019\) P06032](#) | [arXiv:1903.01216](#) | [DOI:10.1088/1748-0221/14/06/P06032](#) | 2019-03-04

678. Search for a heavy pseudoscalar boson decaying to a Z and a Higgs boson at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 564](#) | [arXiv:1903.00941](#) | [DOI:10.1140/epjc/s10052-019-7058-z](#) | 2019-03-03

679. Combinations of single-top-quark production cross-section measurements and $|f_{LVV}|$ determinations at $\sqrt{s}=7$ and 8 TeV with the ATLAS and CMS experiments

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[JHEP 05 \(2019\) 088](#) | [arXiv:1902.07158](#) | [DOI:10.1007/JHEP05\(2019\)088](#) | 2019-02-19

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 773](#) | [arXiv:1902.04374](#) | [DOI:10.1140/epjc/s10052-019-7276-4](#) | 2019-02-12

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[Phys.Lett.B 799 \(2019\) 135049](#) | [arXiv:1902.03603](#) | [DOI:10.1016/j.physletb.2019.135049](#) | 2019-02-10

682. Measurement of exclusive $\rho(770)^0$ photoproduction in ultraperipheral pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 702](#) | [arXiv:1902.01339](#) | [DOI:10.1140/epjc/s10052-019-7202-9](#) | 2019-02-04

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 132001](#) | [arXiv:1902.00571](#) | [DOI:10.1103/PhysRevLett.122.132001](#) | 2019-02-01

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 151802](#) | [arXiv:1901.11201](#) | [DOI:10.1103/PhysRevLett.122.151802](#) | 2019-01-30

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Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 044902](#) | [arXiv:1901.07997](#) | [DOI:10.1103/PhysRevC.100.044902](#) | 2019-01-23

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[Eur.Phys.J.C 79 \(2019\) 444](#) | [arXiv:1901.06726](#) | [DOI:10.1140/epjc/s10052-019-6926-x](#) | 2019-01-20

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Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 281](#) | [arXiv:1901.04060](#) | [DOI:10.1016/j.physletb.2019.05.042](#) | 2019-01-13

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Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 122](#) | [arXiv:1901.03428](#) | [DOI:10.1007/JHEP04\(2019\)122](#) | 2019-01-10

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Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 141](#) | [arXiv:1901.01553](#) | [DOI:10.1007/JHEP03\(2019\)141](#) | 2019-01-06

690. Search for the pair production of light top squarks in the $e^\pm \mu^\mp$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 101](#) | [arXiv:1901.01288](#) | [DOI:10.1007/JHEP03\(2019\)101](#) | 2019-01-04

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Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 112003](#) | [arXiv:1901.00174](#) | [DOI:10.1103/PhysRevD.99.112003](#) | 2019-01-01

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 368](#) | [arXiv:1812.10505](#) | [DOI:10.1140/epjc/s10052-019-6863-8](#) | 2018-12-26

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Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 059](#) | [arXiv:1812.10529](#) | [DOI:10.1007/JHEP12\(2019\)059](#) | 2018-12-26

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Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 313](#) | [arXiv:1812.10534](#) | [DOI:10.1140/epjc/s10052-019-6788-2](#) | 2018-12-26

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[Phys.Lett.B 800 \(2020\) 135042](#) | [arXiv:1812.10514](#) | [DOI:10.1016/j.physletb.2019.135042](#) | 2018-12-26

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 364](#) | [arXiv:1812.09768](#) | [DOI:10.1140/epjc/s10052-019-6855-8](#) | 2018-12-23

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Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 369](#) | [arXiv:1812.06504](#) | [DOI:10.1016/j.physletb.2019.03.059](#) | 2018-12-16

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[Eur.Phys.J.C 79 \(2019\) 208](#) | [arXiv:1812.06489](#) | [DOI:10.1140/epjc/s10052-019-6688-5](#) | 2018-12-16

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[Phys.Lett.B 795 \(2019\) 398](#) | [arXiv:1812.06359](#) | [DOI:10.1016/j.physletb.2019.06.021](#) | 2018-12-15

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Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 031](#) | [arXiv:1812.06302](#) | [DOI:10.1007/JHEP03\(2019\)031](#) | 2018-12-15

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[Eur.Phys.J.C 79 \(2019\) 391](#) | [arXiv:1812.04095](#) | [DOI:10.1140/epjc/s10052-019-6861-x](#) | 2018-12-10

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[JHEP 05 \(2019\) 043](#) | [arXiv:1812.01691](#) | [DOI:10.1007/JHEP05\(2019\)043](#) | 2018-12-04

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[Phys.Lett.B 796 \(2019\) 131](#) | [arXiv:1812.00380](#) | [DOI:10.1016/j.physletb.2019.07.013](#) | 2018-12-02

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[Phys.Rev.Lett. 122 \(2019\) 121803](#) | [arXiv:1811.09689](#) | [DOI:10.1103/PhysRevLett.122.121803](#) | 2018-11-23

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[Eur.Phys.J.C 79 \(2019\) 305](#) | [arXiv:1811.09760](#) | [DOI:10.1140/epjc/s10052-019-6800-x](#) | 2018-11-23

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[Phys.Lett.B 793 \(2019\) 320](#) | [arXiv:1811.08459](#) | [DOI:10.1016/j.physletb.2019.03.064](#) | 2018-11-20

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[Phys.Rev.Lett. 121 \(2018\) 121801](#) | [arXiv:1808.08242](#) | [DOI:10.1103/PhysRevLett.121.121801](#) | 2018-08-24

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[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

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765. Search for supersymmetry in events with a τ lepton pair and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

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Sirunyan, Albert M et al.
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Sirunyan, Albert M et al.
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Sirunyan, Albert M et al.
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804. Search for a heavy right-handed W boson and a heavy neutrino in events with two same-flavor leptons and two jets at $\sqrt{s}= 13$ TeV

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[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

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[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

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[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

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[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

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[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

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[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

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[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

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[Phys.Lett.B 782 \(2018\) 440](#) | [arXiv:1801.01846](#) | [DOI:10.1016/j.physletb.2018.05.062](#) | 2018-01-05

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[JHEP 05 \(2018\) 127](#) | [arXiv:1801.00359](#) | [DOI:10.1007/JHEP05\(2018\)127](#) | 2017-12-31

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[Eur.Phys.J.C 78 \(2018\) 589](#) | [arXiv:1712.09814](#) | [DOI:10.1140/epjc/s10052-018-6049-9](#) | 2017-12-28

832. Search for R-parity violating supersymmetry in pp collisions at $\sqrt{s}=13$ TeV using b jets in a final state with a single lepton, many jets, and high sum of large-radius jet masses

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[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

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[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

835. Search for lepton flavour violating decays of the Higgs boson to $\mu\tau$ and $e\tau$ in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

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[Phys.Rev.Lett. 120 \(2018\) 202005](#) | [arXiv:1712.06144](#) | [DOI:10.1103/PhysRevLett.120.202005](#) | 2017-12-17

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[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30

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Sirunyan, Albert M et al.

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[Phys.Lett.B 776 \(2018\) 355](#) | [arXiv:1705.10141](#) | [DOI:10.1016/j.physletb.2017.11.043](#) | 2017-05-29

922. Combination of searches for heavy resonances decaying to WW, WZ, ZZ, WH, and ZH boson pairs in proton-proton collisions at $\sqrt{s}=8$ and 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 533](#) | [arXiv:1705.09171](#) | [DOI:10.1016/j.physletb.2017.09.083](#) | 2017-05-25

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Sirunyan, Albert M et al.

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925. Search for Charged Higgs Bosons Produced via Vector Boson Fusion and Decaying into a Pair of W and Z Bosons Using pp Collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 141802](#) | [arXiv:1705.02942](#) | [DOI:10.1103/PhysRevLett.119.141802](#) | 2017-05-08

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 746](#) | [arXiv:1705.02628](#) | [DOI:10.1140/epjc/s10052-017-5286-7](#) | 2017-05-07

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[Phys.Lett.B 774 \(2017\) 279](#) | [arXiv:1705.01403](#) | [DOI:10.1016/j.physletb.2017.09.053](#) | 2017-05-03

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[Phys.Rev.D 96 \(2017\) 032002](#) | [arXiv:1704.06142](#) | [DOI:10.1103/PhysRevD.96.032002](#) | 2017-04-20

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Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 001](#) | [arXiv:1704.03366](#) | [DOI:10.1007/JHEP07\(2017\)001](#) | 2017-04-11

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Sirunyan, A.M. et al.

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Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

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[Phys.Lett.B 772 \(2017\) 21](#) | [arXiv:1703.06095](#) | [DOI:10.1016/j.physletb.2017.06.009](#) | 2017-03-17

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 459](#) | [arXiv:1703.01630](#) | [DOI:10.1140/epjc/s10052-017-4984-5](#) | 2017-03-05

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942. Measurement of the cross section for electroweak production of $Z\gamma$ in association with two jets and constraints on anomalous quartic gauge couplings in proton-proton collisions

at $\sqrt{s} = 8$ TeV

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[Phys.Lett.B 770 \(2017\) 380](#) | [arXiv:1702.03025](#) | [DOI:10.1016/j.physletb.2017.04.071](#) | 2017-02-09

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[Eur.Phys.J.C 77 \(2017\) 269](#) | [arXiv:1702.01462](#) | [DOI:10.1140/epjc/s10052-017-4828-3](#) | 2017-02-05

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[Phys.Rev.Lett. 119 \(2017\) 082301](#) | [arXiv:1702.01060](#) | [DOI:10.1103/PhysRevLett.119.082301](#) | 2017-02-03

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[Phys.Lett.B 776 \(2018\) 195](#) | [arXiv:1702.00630](#) | [DOI:10.1016/j.physletb.2017.11.041](#) | 2017-02-02

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[Phys.Lett.B 772 \(2017\) 634](#) | [arXiv:1701.08328](#) | [DOI:10.1016/j.physletb.2017.07.022](#) | 2017-01-28

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[JHEP 09 \(2017\) 051](#) | [arXiv:1701.06228](#) | [DOI:10.1007/JHEP09\(2017\)051](#) | 2017-01-22

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[Phys.Rev.D 96 \(2017\) 012004](#) | [arXiv:1701.01954](#) | [DOI:10.1103/PhysRevD.96.012004](#) | 2017-01-08

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[Phys.Lett.B 773 \(2017\) 563](#) | [arXiv:1701.01345](#) | [DOI:10.1016/j.physletb.2017.08.069](#) | 2017-01-05

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[Phys.Lett.B 772 \(2017\) 363](#) | [arXiv:1612.09516](#) | [DOI:10.1016/j.physletb.2017.06.062](#) | 2016-12-30

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[Phys.Lett.B 772 \(2017\) 306](#) | [arXiv:1612.08972](#) | [DOI:10.1016/j.physletb.2017.06.053](#) | 2016-12-28

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[Eur.Phys.J.C 77 \(2017\) 92](#) | [arXiv:1608.07561](#) | [DOI:10.1140/epjc/s10052-016-4573-z](#) | 2016-08-26

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Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 123](#) | [arXiv:1608.03560](#) | [DOI:10.1007/JHEP12\(2016\)123](#) | 2016-08-11

1004. Search for new phenomena in events with high jet multiplicity and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 257](#) | [arXiv:1608.01224](#) | [DOI:10.1016/j.physletb.2017.01.073](#) | 2016-08-03

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Khachatryan, Vardan et al.

[Phys.Lett.B 763 \(2016\) 280](#) | [arXiv:1607.08834](#) | [DOI:10.1016/j.physletb.2016.10.054](#) | 2016-07-29

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Phys.Lett.B 766 \(2017\) 268](#) | [arXiv:1607.06943](#) | [DOI:10.1016/j.physletb.2017.01.011](#) | 2016-07-23

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[JHEP 12 \(2016\) 083](#) | [arXiv:1607.05764](#) | [DOI:10.1007/JHEP12\(2016\)083](#) | 2016-07-19

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Khachatryan, Vardan et al.

[Phys.Lett.B 764 \(2017\) 66](#) | [arXiv:1607.02638](#) | [DOI:10.1016/j.physletb.2016.11.001](#) | 2016-07-09

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Phys.Rev.Lett. 117 \(2016\) 051802](#) | [arXiv:1606.04093](#) | [DOI:10.1103/PhysRevLett.117.051802](#) | 2016-06-13

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Phys.Lett.B 762 \(2016\) 512](#) | [arXiv:1605.09047](#) | [DOI:10.1016/j.physletb.2016.10.007](#) | 2016-05-29

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[JHEP 08 \(2016\) 119](#) | [arXiv:1604.04464](#) | [DOI:10.1007/JHEP08\(2016\)119](#) | 2016-04-15

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[JHEP 12 \(2016\) 088](#) | [arXiv:1603.08914](#) | [DOI:10.1007/JHEP12\(2016\)088](#) | 2016-03-29

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[Phys.Lett.B 759 \(2016\) 369](#) | [arXiv:1603.02991](#) | [DOI:10.1016/j.physletb.2016.05.087](#) | 2016-03-09

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Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 169](#) | [arXiv:1603.02248](#) | [DOI:10.1007/JHEP04\(2016\)169](#) | 2016-03-07

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[JHEP 08 \(2016\) 029](#) | [arXiv:1603.02303](#) | [DOI:10.1007/JHEP08\(2016\)029](#) | 2016-03-07

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[Eur.Phys.J.C 76 \(2016\) 469](#) | [arXiv:1603.01803](#) | [DOI:10.1140/epjc/s10052-016-4293-4](#) | 2016-03-06

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[Eur.Phys.J.C 76 \(2016\) 460](#) | [arXiv:1603.00765](#) | [DOI:10.1140/epjc/s10052-016-4292-5](#) | 2016-03-02

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[Eur.Phys.J.C 77 \(2017\) 15](#) | [arXiv:1602.09024](#) | [DOI:10.1140/epjc/s10052-016-4504-z](#) | 2016-02-29

1044. Search for heavy resonances decaying to two Higgs bosons in final states containing four b quarks

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 371](#) | [arXiv:1602.08762](#) | [DOI:10.1140/epjc/s10052-016-4206-6](#) | 2016-02-28

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Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 479](#) | [arXiv:1602.08772](#) | [DOI:10.1016/j.physletb.2016.05.088](#) | 2016-02-28

1046. Measurement of the $Z \gamma \rightarrow \nu \bar{\nu} \gamma$ production cross section in pp collisions at $\sqrt{s}=8$ TeV and limits on anomalous ZZ γ and $Z\gamma\gamma$ trilinear gauge boson couplings

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1047. Search for supersymmetry in the multijet and missing transverse momentum final state in pp collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 152](#) | [arXiv:1602.06581](#) | [DOI:10.1016/j.physletb.2016.05.002](#) | 2016-02-21

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Khachatryan, Vardan et al.

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[Eur.Phys.J.C 76 \(2016\) 325](#) | [arXiv:1601.04768](#) | [DOI:10.1140/epjc/s10052-016-4156-z](#) | 2016-01-18

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Khachatryan, Vardan et al.
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[Phys.Lett.B 733 \(2014\) 328](#) | [arXiv:1311.4937](#) | [DOI:10.1016/j.physletb.2014.04.023](#) | 2013-11-19

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